

a Observed evidence**b** Defensible field claims

Major chemistry	✓	Data readiness / QC	✓
Stable water isotopes	✓	Seasonal chemistry prediction	✓
Sr + SiO ₂	✓	Reaction equivalence classes	✓
Single static-water field	✓	Alternative-edge sensitivity	✓
Environmental age tracers	—	Residence-time validation	—
Screen intervals	—	Exact directed topology	—
Repeated heads	—	Unique field reactions	—
Independent flow truth	—	Validated field digital twin	—

